

Mapping AI's Industry Impact: Evidence from News Data

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Research Objective & Motivation

Objective

This project examines how artificial intelligence may shape industries through large-scale analysis of news data. Using NLP techniques, I aim to identify affected industries, understand impact mechanisms, and derive data-driven insights into AI adoption.

Why This Matters

Artificial intelligence is widely discussed as a transformative force across industries, yet systematic understanding of how these impacts emerge in real-world discourse remains limited.

News articles provide large-scale, continuously updated narratives on AI impacts. Analyzing this web-crawled news data enables data-driven investigation of how AI impact is communicated, perceived, and contextualized across different sectors.

Research Questions

- Which industries and key companies are most likely to be impacted by AI?
- In what direction and through what mechanisms does AI-driven impact occur?
- What factors appear to influence the success or challenges of AI adoption?

Key Challenges & Analytical Work Units

Challenge 1

Noisy Web-Crawled News Content

Web-crawled articles contain artifacts and irrelevant structures that distort semantic analysis.



**Text Cleaning
&
Structural Filtering**

Challenge 2

Identify Industries Impacted by AI

AI impact signals are unevenly distributed across sectors and must be inferred from recurring discourse patterns.



**Topic Modeling
&
Industry Mapping**

Challenge 3

Isolate AI Impact Signal

Industry-related content may still include general or non-impact narratives.



**Industry-Specific
Topic Modeling & Filtering**

Challenge 4

Measure Direction of AI Impact

Understanding whether AI impact is positive, negative, or ambiguous requires robust sentiment modeling.



Sentiment Modeling Pipeline

Challenge 5

Understand Mechanisms of Impact

Understanding whether AI impact must be interpreted through entities such as companies, technologies, and mechanisms.



**Entity Identification
&
Entity-Level Impact Analysis**

Challenge 6

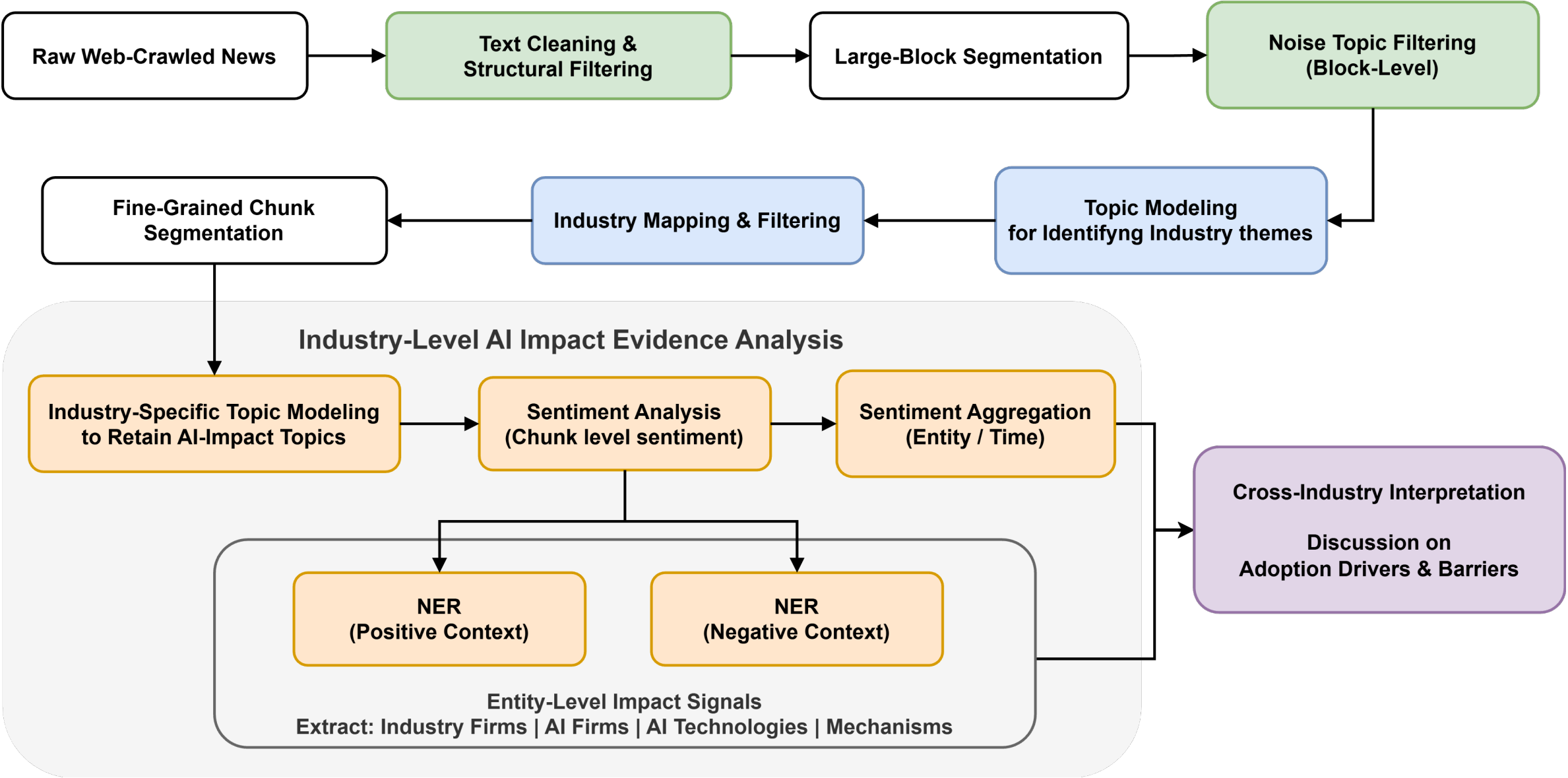
Explain Success & Barriers to AI Adoption

Variation in impact patterns suggests differing conditions for successful adoption across industries.



Interpretive Analysis

Methodological Workflow Overview



Text Cleaning – Data Sanity Checks

Dataset Overview

- ~200K web-crawled news articles on data science and artificial intelligence
- Fields: URL | Date | Language | Title | Text
- Language: English only

Data Quality Diagnostics

- Empty text ratio: 0.0%
- Duplicate text ratio: 0.002%
- Duplicate title ratio: 0.178%
- Text length (characters):
 - Max: 580K
 - Min: 21
 - Mean: 9K

Step 0 Cleaning

1. Deduplication Strategy

- Remove exact duplicate texts
- Remove records with duplicate titles
 - Identical-title articles were examined due to frequent cross-site reprints.
 - Content similarity was validated using TF-IDF cosine similarity.
 - Most duplicated-title articles share near-identical main content.
- Retain earliest instance

2. Text Normalization

- Normalize whitespace
- Unicode standardization

Text Cleaning – Step 1: Trim Pre-Title Web Artifacts

Observed Crawl Pattern

Scraped news pages often contain duplicated titles and webpage artifacts:

[Title]
[Navigation / menus / share buttons / site content]
[Title]
[Actual article content...]
[buttons / site content]

Cleaning Rule

- The second occurrence of the title is treated as the true beginning of the article body.
- Content appearing before this point is removed as webpage noise.

Note : This rule-based filtering removes many template artifacts but does not fully eliminate all structural noise. Remaining noise is addressed in later filtering stages.

Title repetition statistics confirm that multiple title occurrences are common in scraped pages.

title_occur_counts	
1	107897
2	28093
3	11503
0	10356
4	4847
5	1239

3 Examples of Removed Web Artifacts

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say\n×\n\nBaseball\n\nNFL\n\nNBA\n\nNHL\n\nWWE\n\nNCAAF\n\nWNBA\n\nGolf\n\nTennis\n\nSoccer\n\nMMA\n\nNCAAB\n\nGaming\n\nAEW\n\nFortnite\n\nMinecraft\n\n\n\nManage your profile\n\nLogout\n\nBaseball\n\nMLB\n\nSchedule\n\nTop MLB Picks\n\nPredictions\n\nNew York Yankees\n\nShohei Ohtani\n\nAlex Rodriguez

We asked AI

141576

The Global Race To Build The AI University: Can Anyone Catch Up With China?\n\nFree Press Journal\n\nPaper\n\nGet App\n\nHome\n\nMumbai News

ReviewsBusinessTechnologyGamesFeaturesV

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(US)\n\nArmenia\n\nAustralia\n\nAustria\n\nAzerbaijan\n\nBahamas\n\nBahrain\n\nBangladesh\n\nBarbados\n\nBelarus\n\nBelgium\n\nBolivia\n\nBotswana\n\nBrazil\n\nBrunei\n\nBulgaria\n\nBurkina Faso\n\nBurundi\n\nCalifornia\n\nCambodia\n\nCameroon\n\nCanada\n\nCape Verde\n\nCayman Islands\n\nCzechia\n\nCongo\n\nCongo-Brazzaville\n\nCongo-Kinshasa\n\nConnecticut\n\nCosta Rica\n\nCroatia\n\nCuba\n\nCyprus\n\nCzech Republic\n\nDelaware\n\nDjibouti\n\nDominica\n\nDominican Republic\n\nEcuador\n\nEgypt\n\nEl Salvador\n\nEquatorial Guinea\n\nEritrea\n\nEstonia\n\nEthiopia\n\nFalkland Islands\n\nFiji\n\nFinland\n\nFrance\n\nGermany\n\nGhana\n\nGibraltar\n\nGreece\n\nGuam\n\nGuatemala\n\nGuinea\n\nGuinea-Bissau\n\nGuyana\n\nHaiti\n\nHawaii\n\nHonduras\n\nHungary\n\nIndonesia\n\nIowa\n\nIran\n\nIraq\n\nIsrael\n\nItaly\n\nIvory Coast\n\nJapan\n\nJordan\n\nKenya\n\nKosovo\n\nKuwait\n\nKyrgyzstan\n\nLaos\n\nLatvia\n\nLebanon\n\nLesotho\n\nLiberia\n\nLibya\n\nLiechtenstein\n\nLithuania\n\nLuxembourg\n\nMalawi\n\nMalaysia\n\nMali\n\nMalta\n\nMaryland\n\nMassachusetts\n\nMauritania\n\nMauritius\n\nMexico\n\nMichigan\n\nMinnesota\n\nMoldova\n\nMonaco\n\nMontenegro\n\nMorocco\n\nMozambique\n\nMyanmar\n\nNamibia\n\nNebraska\n\nNepal\n\nNetherlands\n\nNevada\n\nNew Caledonia\n\nNew Zealand\n\nNicaragua\n\nNiger\n\nNigeria\n\nNorth Carolina\n\nNorth Dakota\n\nNorth Korea\n\nNorth Macedonia\n\nNorway\n\nOhio\n\nOklahoma\n\nOman\n\nPakistan\n\nPanama\n\nParaguay\n\nPennsylvania\n\nPeru\n\nPhilippines\n\nPoland\n\nPortugal\n\nPuerto Rico\n\nQatar\n\nReunion\n\nRhode Island\n\nRomania\n\nRussia\n\nSenegal\n\nSerbia\n\nSeychelles\n\nSierra Leone\n\nSingapore\n\nSlovakia\n\nSlovenia\n\nSomalia\n\nSouth Africa\n\nSouth Korea\n\nSouth Sudan\n\nSudan\n\nSuriname\n\nSwaziland\n\nSweden\n\nSwitzerland\n\nSyria\n\nTaiwan\n\nTajikistan\n\nTanzania\n\nTennessee\n\nTexas\n\nThailand\n\nTimor-Leste\n\nTogo\n\nTonga\n\nTrinidad and Tobago\n\nTunisia\n\nTurkey\n\nUganda\n\nUkraine\n\nUnited Arab Emirates\n\nUnited Kingdom\n\nUnited States\n\nUruguay\n\nUtah\n\nUzbekistan\n\nVanuatu\n\nVatican City\n\nVenezuela\n\nVermont\n\nVietnam\n\nVirginia\n\nVirgin Islands\n\nWashington\n\nWest Virginia\n\nWisconsin\n\nWyoming\n\nYemen\n\nZambia\n\nZimbabwe

Newswires by US State\n\nAlabama\n\nAlaska\n\nArizona\n\nArkansas\n\nCalifornia\n\nColorado\n\nConnecticut\n\nDelaware\n\nFlorida\n\nGeorgia\n\nHawaii\n\nIdaho\n\nIllinois\n\nIndiana\n\nIowa\n\nKansas\n\nKentucky\n\nLouisiana\n\nMaine\n\nMaryland\n\nMassachusetts\n\nMichigan\n\nMinnesota\n\nMississippi\n\nMissouri\n\nMontana\n\nNebraska\n\nNevada\n\nNew Hampshire\n\nNew Jersey\n\nNew Mexico\n\nNew York\n\nNorth Carolina\n\nNorth Dakota\n\nOhio\n\nOklahoma\n\nOregon\n\nPennsylvania\n\nRhode Island\n\nSouth Carolina\n\nSouth Dakota\n\nTennessee\n\nTexas\n\nUtah\n\nVermont\n\nVirginia\n\nWashington\n\nWest Virginia\n\nWisconsin\n\nWyoming

Live Feed\n\nKnowledge Base\n\nEditorial Guid

Press Release\n\nFAQs\n\nRSS Feeds\n\nEmail Newsletters\n\nNews Alert Maker\n\nAffiliate Program\n\nQuestions\n\n+1 (202) 335-3939\n\nPricing\n\n&\n\nComparison

World Media Directory\n\nRSS\n\nEmail Newsletters\n\nThere were 17

Text Cleaning – Step 2: Document-Level Heuristic Filtering

1. Remove emoji

- Emoji characters are removed using regex-based filtering.
- They do not contribute to semantic analysis of AI impact.

2. Remove Title-Dominant Documents

- Title–text similarity is computed to detect documents lacking substantive body content.
- Similarity evaluation is restricted to short documents for efficiency.
- High similarity or exact matches indicate title-only pages, which are removed.

3. Remove based on URL

Domains and paths associated with non-article or low-content pages are excluded.

• **Domain-level exclusions (non-news / low-content):**

- rawpixel.com → image-only pages
- levels.fyi → job listings salary pages
- emoji.fly.dev → emoji metadata
- collider.com → entertainment content
- 365project.org → non-news pages
- hackernoon.com → signup / productized story pages

• **Path-level exclusions (pattern-based):**

- mexc.* /price/ – crypto price pages
- mexc.* /exchange/ – exchange listings
- wionews.com/videos/ → video hubs

Text Cleaning – Step 3: Structural Artifact & Noise Removal

Web-crawled news pages contain template-driven structural elements that can distort topic modeling and downstream semantic analysis. This step focuses on removing template-level structural noise using regex.

HTML Artifacts

- Remove residual HTML fragments (e.g., tags, layout markers) to ensure clean textual representation.

Short Runs from Blocks

- Discard very short text fragments generated during block segmentation (e.g., navigation labels, button text).
- Preserve short runs containing AI-related keywords to avoid false deletions.
- This step reduces, but does not fully eliminate, boilerplate content.

Duplication Blocks

- Remove repeated structural sections such as headers, footers, and “related articles” modules.

Sign-In Specific Content

Login prompts, subscription banners, and account notices

Cookies & Privacy Policies

Cookie notices and privacy policy fragments

Language / Country Selection Elements

Language switchers and region selectors

Text Cleaning – Step 4: Final Filtering & Clean Corpus Summary

Final Keyword Retention Rule

Documents are retained when BOTH conditions hold:

- AI-related terms appear in the text
- Impact signals appear in the text
OR AI is explicitly mentioned in the title

Extreme Length Filtering

- Manually inspect documents with length >150K or <800 characters
- Remove extremely long irrelevant pages and very short low-content texts

Whitespace Normalization

Multiple line breaks and redundant spaces may arise during preprocessing

- Consecutive line breaks are normalized to paragraph breaks (\n\n)
- Excessive spaces are collapsed into a single space

Data Cleaning Summary

- Number of entries: 149K
- Empty text ratio: 0.0%
- Duplicate text ratio: 0.0%
- Duplicate title ratio: 0.0%
- Text length (characters):
 - Max: 227K
 - Min: 800
 - Mean: 7.2K
 - Median: 6K

Identify Industries – Step 1: Text Splitting & Noise Topic Filtering

Motivation for Text Segmentation

- Transformer embedding models (e.g., BERT) impose a **~512-token limit**. Directly truncating texts may result in substantial information loss.
- Allow residual boilerplate or structural webpage fragments to emerge as separate units

Hierarchical text segmentation strategy

Long documents (> 6,000 characters) are split progressively:

1. Paragraph-level splitting: using double line breaks (\n\n) to preserve major discourse structure
2. Line-level splitting: using single line breaks (\n) when blocks remain too long
3. Sentence-level splitting: as a final fallback for excessively long segments

This hierarchical segmentation ensures that resulting blocks fall within **4K–6K character range**, balancing semantic integrity and modeling feasibility.

Fine-grained topic clustering for noise filtering

Following segmentation, a first-stage semantic filtering is performed.

- The clustering process produces **~290 topics**, capturing both meaningful content and structural noise
- Topics whose keyword representations contain **no AI-related words** are manually inspected whether they are boilerplates
- Boilerplate clusters are removed
- Topic -1 is retained to avoid discarding potentially meaningful but sparse content

This step functions as **semantic noise filtering**, enabling subsequent modeling to focus on substantive AI-related narrative signals.

Identify Industries – Step 2: Industry Mapping & Post Processing

Coarser Topic Modeling for Industry Signals

After removing structural noise topics, a second round of coarser topic modeling is applied.

- A second, coarser topic modeling stage is applied after noise filtering
- Produces **~170 topics**, capturing higher-level semantic structures
- Focus shifts from noise removal to **latent industry theme discovery**



Multi-representation Keyword Inspection:

- To improve interpretability, multiple CountVectorizer configurations are used to generate alternative keyword representations:
- Different max_df, min_df, and n-gram settings produce varied views of topic semantics
- Domain-specific and AI-related stopwords are incorporated to reduce generic signal.

LLM-assisted industry classification

- Topics are classified into **10 predefined industry categories**
- **Deepseek LLM API*** is used to perform industry classification.
 - Keyword representations serve as the primary evidence
 - Representative documents provide conservative validation
- Topics unrelated to the industry are **dropped**

Industry taxonomy is informed by:

- NAICS(Government)-inspired sector definitions
- Manual semantic inspection



Post-processing: Fine-grained Chunking

- Blocks are further split into ~12-sentence windows
- Improves topic purity and enables downstream sentiment & NER analysis

* For LLM-based classification, temperature is set to 0 to ensure deterministic and reproducible outputs.

Identify Industries – 10 Predefined Industry Categories



01. Physical Production

Agriculture, Mining,
Manufacturing, Energy



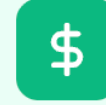
02. Infrastructure Systems

Utilities, Construction,
Transport, Smart Cities



03. Consumer Economy

Retail, E-Commerce,
Food & Accommodation



04. Financial Systems

Capital Markets, FinTech,
Insurance, Real Estate



05. Healthcare Systems

Health Services, Health
Tech, Medical Devices



06. Education & Research

Academia, Knowledge
Products, Training



07. Media & Creative

Content, News, Film,
Gaming, Social Media



08. Digital Services

Software, AI, Cloud,
Data, CyberSec



09. Public Sector

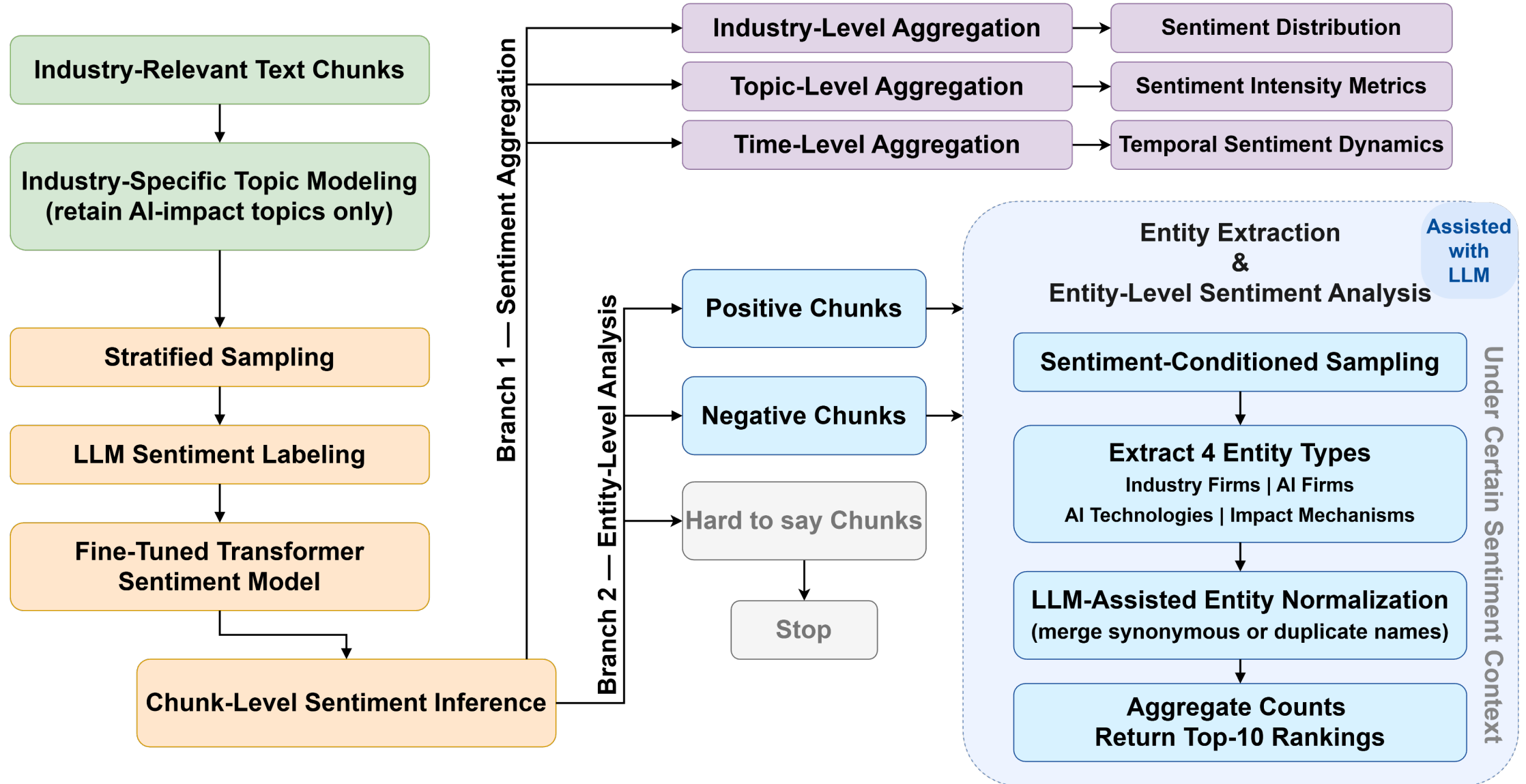
Gov, NGOs, Civic Orgs,
Public Services



10. Other

Unmapped industries
and miscellaneous

Industry Analysis – Sentiment and NER Pipeline Overview



Industry Analysis – Industry Specific Pipeline Variations

- Different industries exhibit distinct impact patterns of AI
- Therefore analytical strategies are adjusted accordingly

Financial Systems

- A large portion of articles discusses fluctuations in AI-related stocks and financial assets
- **Additional sentiment analysis** is conducted specifically on AI stock signals (*positive vs. non-positive*)

Physical Production

- AI affects high-tech manufacturing and traditional production through different mechanisms
- These **sub-sectors are analyzed separately** to capture heterogeneous impact pathways

Information & Digital Services

- Rather than being impacted, firms in this sector often act as primary AI impact drivers
- **Sentiment analysis is therefore deprioritized**, with focus placed on entity-level extraction (*companies, AI products, technologies, impact mechanisms*)

Industry Analysis – Step1: Industry Specific Topic Modeling

Coarse Topic Modeling within Each Industry

- BERTopic is applied again within each industry using a **coarser clustering setting**
- UMAP n_neighbors is increased; HDBSCAN min_cluster_size and min_samples are increased
- UMAP n_components is reduced to encourage broader semantic grouping
- This typically yields **~5–20 high-level topics** per industry

Identify AI-Impact-Relevant Topics

- Topic keywords and representative documents are reviewed to determine **whether** the topic is related to **AI impact** within that industry (*AI Impact / General or Not Related*)
- Topic relevance is assessed through:
 - manual inspection of keywords / representative texts
 - or LLM*-assisted semantic review
- **Industry-Specific Narrative Identification**
 - Financial Systems: Topics distinguish between AI impact topics and AI stock topics
 - Physical Production: Topics distinguish between AI in high-tech manufacturing and AI in traditional production sectors

Filter Irrelevant Topics

- Topics discussing general industry content but not **AI impact** are removed
- Only AI-impact-relevant topics are retained for downstream sentiment and entity analysis

* For LLM-based classification, temperature is set to 0 to ensure deterministic and reproducible outputs.

Industry Analysis – Step2: Industry-Aware Sentiment Modeling

1. Topic-stratified sampling (~500–600 chunks)

- Ensures minority topics receive sufficient representation
- Mitigates sampling bias in downstream sentiment modeling
- Preserves topic-level semantic diversity in training data

2. LLM-assisted weak labeling for sentiment initialization

- 3 sentiment classes: positively / negatively / hard to say (or irrelevant)
- Provides scalable labeling for domain-specific discourse
- LLM-guided* augmentation applied when lacking sufficient negative samples
 - Reduces model bias toward dominant sentiment classes
 - Ensures minimum class balance (~60 negative samples per topic)

3. Fine-tune a transformer-based sentiment classifier (DistilBERT)

- Selected for efficiency–accuracy trade-off in large-scale inference
- Macro-F1 used for model selection, emphasizing balanced performance under class imbalance

4. Final model applied to full corpus for chunk-level sentiment inference

* Unlike classification tasks (temperature = 0), text augmentation uses temperature = 1.1 to introduce stylistic and lexical diversity in generated samples.

Industry Analysis – Step3: Sentiment Aggregation

- Chunk-level sentiment predictions are aggregated across **three analytical levels** to interpret AI impact signals.
- Aggregation uses **unique document counts per sentiment label** (instead of chunk counts) to reduce length bias.

Industry-Level Aggregation

- Compute sentiment distribution across all industry-relevant documents
- Report:
 - sentiment ratio (positive / neutral / negative)
- Used for:
 - macro-level interpretation of AI impact direction
 - cross-industry comparison

Time-Level Aggregation

- Temporal sentiment dynamics are analyzed at the year–quarter level
- Two visualization forms are used:
 - Absolute sentiment volume (stacked area)
 - Relative sentiment proportion (stacked area)
- This enables:
 - identification of sentiment trend shifts
 - alignment with real-world AI adoption events

Topic-Level Aggregation

- Topics are first semantically summarized using LLM.
- When topic granularity is excessive, topics are grouped into 3–5 meta-topics via:
 - manual inspection
 - or LLM-assisted semantic clustering
- Weighted sentiment score: $Score = P_{positive} - P_{negative}$
 - where: positive = +1, hard-to-say = 0, negative = -1
- **Bias-Corrected Sentiment Estimation**
 - Fine-tuned classifier predictions exhibit systematic bias (diagnosed via confusion matrix).
 - a **distribution correction step** is applied:
 - Treat predicted label distribution as a noisy transformation of true labels
 - Apply inverse confusion matrix transformation
 - Obtain **adjusted sentiment score**
 - This improves:
 - comparability across topics
 - robustness of downstream interpretation

Industry Analysis – Step4: Sentiment-Level NER

1 Sentiment-Conditioned Entity Extraction

- Text chunks are first separated by sentiment label (positive vs. negative AI impact).
- Sampling before NER tasks to control cost:
 - ~800 positive samples
 - ~500 negative samples
- Extraction prompts are **sentiment-aware**
- A structured LLM extraction pipeline identifies four entity types:
 - Industry firms
 - AI firms
 - AI technologies
 - Impact mechanisms

2 Entity Aggregation & Normalization

- Extracted entities are first expanded into (*entity*, *count*) pairs.
- Pairs with count > 1 are passed to an **LLM-assisted normalization step** to:
 - merge synonymous or duplicate names
 - standardize entity representations
 - aggregate mention frequencies
- Final outputs include **Top-10 ranked entities per sentiment context**.

Analytical Framing: Sentiment-Conditioned Entity Frequency

- My approach estimates: $P(\textit{entity} \mid \textit{sentiment})$
 - Measures which companies and mechanisms dominate positive vs. negative AI narratives.
 - Represents a **narrative-composition perspective** rather than reputational evaluation.
- Compared to the alternative: $P(\textit{sentiment} \mid \textit{entity})$
 - Focuses on sentiment attribution toward specific entities.
 - Requires entity-level sentiment aggregation across heterogeneous contexts, which is limited in this dataset.

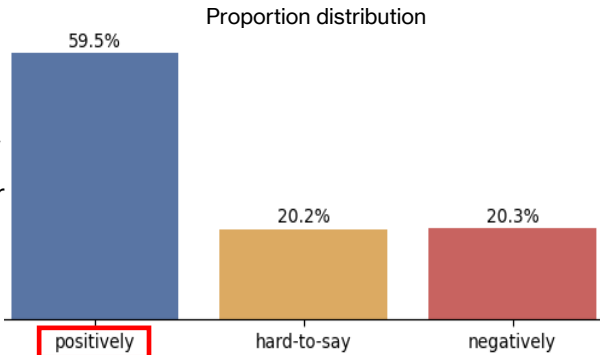
Rationale for This Design

- More suitable for the project objective → understanding **how AI impact discourse is constructed**.
- Advantages:
 - Directly reflects narrative framing structures
 - More stable under sparse entity mentions
 - Avoids multi-entity sentiment attribution ambiguity and easier to implement in this dataset
 - Easier interpretation in policy and industry analysis contexts

Industry Analysis Result – Infrastructure Systems Industry

Industry Level Sentiment

Infrastructure AI discussions are **largely positive** but **increasingly** shaped by **concerns** over capacity constraints, and energy demand concern.

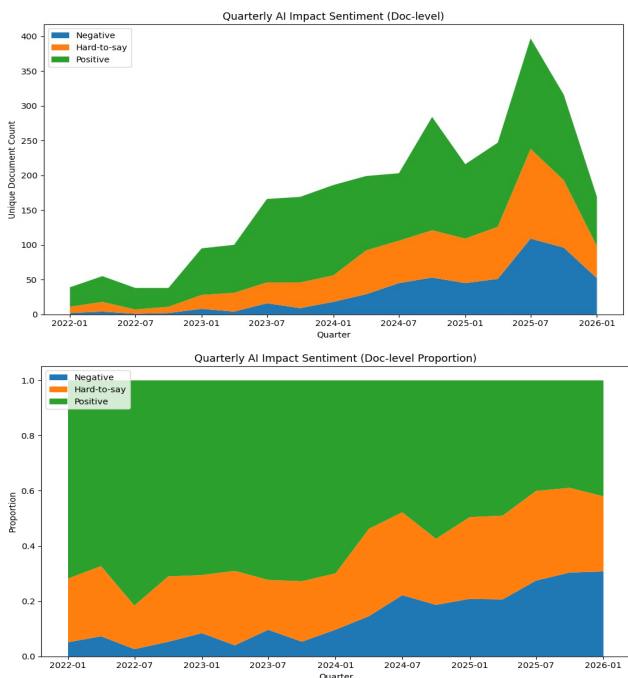


Topic – Level Sentiment

Topic	Sent Score	Ad. Sent Score
Data centers' high water consumption and emissions raise environmental concerns amid climate change	0.25	0.03
The evolution of wireless telecom networks, including 5G, AI integration, and infrastructure from major vendors.	0.70	0.51
Using artificial intelligence to reduce aviation's climate impact	0.67	0.76

Time Level Sentiment

- As AI adoption in infrastructure scales up
- Discourse shifts from optimism toward a more balanced narrative
- Negative discourse is increasing over time



NER in Positive content

- Infrastructure companies: Vodafone, Huawei, Nokia, Ericsson, Constellation Energy
- AI companies: Microsoft, Google, OpenAI, Nvidia, IBM
- AI Technologies: machine learning, generative AI, large language models, predictive analytics, digital twins, computer vision
- Positive mechanisms: predictive maintenance, energy efficiency optimization automated infrastructure management, power demand forecasting, logistics optimization

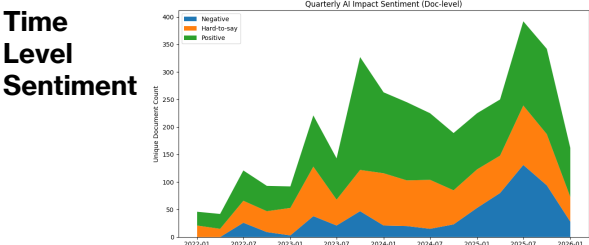
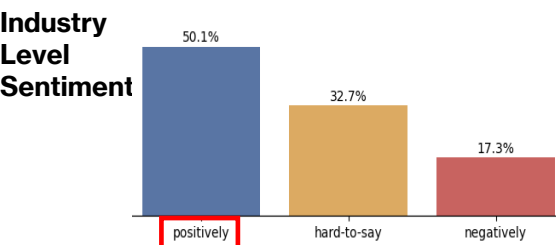
NER in Negative content

- Infrastructure companies: PJM Interconnection, Constellation Energy, Dominion Energy, Georgia Power, PG&E
- AI companies: Microsoft, Google, Amazon, OpenAI, Meta
- AI Technologies: generative AI, large language models, machine learning, cloud computing
- Negative mechanisms: infrastructure capacity shortages, data center power consumption concerns, power grid stress, infrastructure investment pressure

Industry Analysis Result – Physical Production Industry

High-Tech Production

AI impact is **predominantly opportunity-driven**, with negative narratives concentrated around **geopolitical chip restrictions**.



Topic – Level Sentiment

Topic	Sent Score	Ad. Sent Score
Digital innovation within the automotive industry	0.47	0.44
U.S. export controls on advanced AI chips to China over national security concerns.	-0.40	-0.78
Taiwan Semiconductor Manufacturing	0.57	0.64
Tesla's humanoid robot and its self-driving technology	0.57	0.62
AI memory chip	0.77	0.86
Google's new Pixel smartphones	0.67	0.78
Intel's new AI-focused chips	0.99	1.00

NER in Positive content

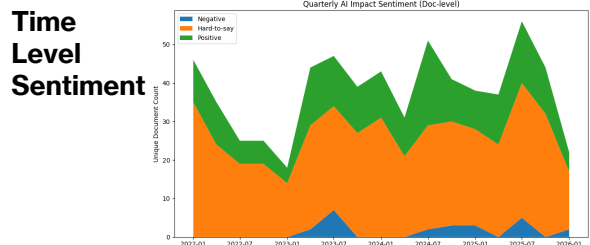
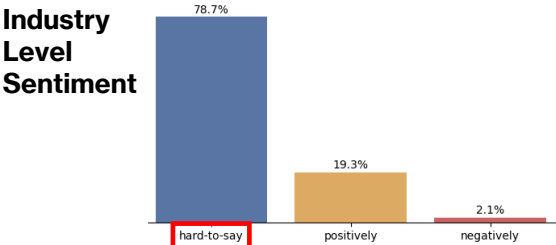
- High-tech companies: Nvidia, Intel, TSMC, Samsung, AMD
- AI companies: OpenAI, Google, Microsoft, Amazon, Meta
- AI Products: AI Chips, GPUs, HBMs, AI Processor, NPUs, Autonomous driving
- Positive mechanisms: demand and revenue growth, capacity expansion needed, efficiency, AI applications

NER in Negative content (mostly chip export control texts)

- High-tech companies: Nvidia, AMD, Huawei
- AI companies: Alibaba, Tencent, Bytedance, DeepSeek
- AI Products: AI Chips, GPUs, Semiconductors
- Negative mechanisms: export restrictions, supply chain disruption, technological disadvantage, competitive pressure, manufacturing delays

Traditional Production

Predominantly **neutral**, with positive narratives focused on **efficiency gains** and negative narratives centered on **labor, implementation costs and risks**.



Topic – Level Sentiment

Topic	Sent Score	Ad. Sent Score
Agriculture and Farmers	0.25	0.35
Drilling and Exploration	0.27	0.36

NER in Positive content

- Traditional companies: John Deere, CNH Industrial, Case IH Agriculture, New Holland
- AI companies: Cropin, IBM, Microsoft, FarmERP
- AI technologies: AI, predictive analysis, computer vision, automation
- Positive mechanisms: precision farming, crop monitoring, yield improvement, production optimization, resource allocation

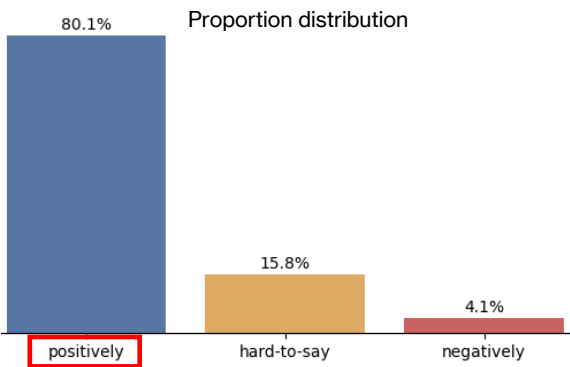
NER in Negative content

- Traditional companies: International Federation of Robotics, BharatAgri, Evogene, Digital Agriculture Lab
- AI technologies: AI, predictive analysis, drones,
- Negative mechanisms: job loss due to automation, high initial costs, labor costs, skill gaps, decreased reliability, struggled to consistently provide robust results, trust in forecasts

Industry Analysis Result – Consumer Economy Industry

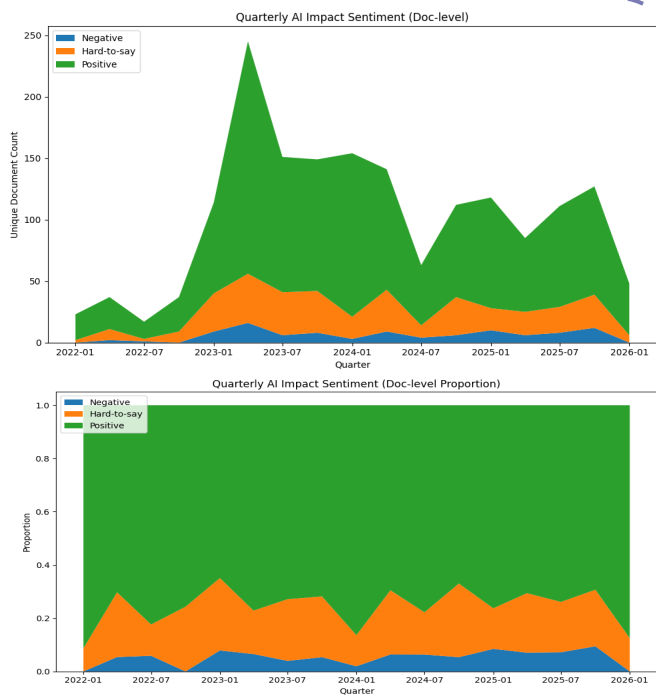
Industry Level Sentiment

Overwhelmingly framed **positively**, enhancing **customer experience and operational efficiency**, while negative discourse primarily focused on **implementation risks**.



Time Level Sentiment

- AI discourse in the consumer economy remains strongly positive



In time analysis plot: ● Positive, ● Hard to say, ● Negative

Topic – Level Sentiment

Topic	Sent Score	Ad. Sent Score
Using technology like AI to improve both online and in-store retail shopping experiences	0.80	0.76
Fast-food restaurants are testing AI chatbot to automate drive-thru orders	0.67	0.59
AI and apps for travel planning and itinerary creation	0.73	0.68

NER in Positive content

- Consumer economy companies: Walmart, Amazon, Wendy’s, McDonald’s, Expedia, Yum! Brands, Airbnb
- AI companies: Google, Microsoft, OpenAI, IBM
- AI Technologies: generative AI, large language models, computer vision, natural language processing, predictive analytics, conversational AI, voice AI, robotics
- Positive mechanisms: chatbots, personalized recommendations, AI assistants, inventory management, AI-powered customer service, workflow automation, AI-powered search, dynamic pricing, autonomous checkout, demand forecasting

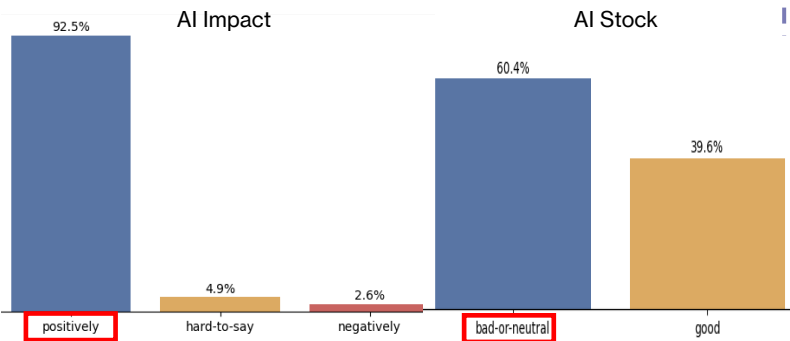
NER in Negative content

- Consumer economy companies: McDonald’s, Walmart, Instacart, Wendy’s, Taco Bell, Starbucks
- AI companies: Google, TikTok, Microsoft, IBM, Chef Robotics, Veesion
- AI Technologies: generative AI, machine learning, robotics, computer vision, self-checkout kiosks, big data, voice ordering technology
- Negative mechanisms: AI hallucinations, AI distortions, AI pricing manipulation, AI drive-thru ordering errors, biased AI content, feedback loops in AI systems, AI’s limitations in economic understanding, AI lacking accountability

Industry Analysis Result – Financial Systems Industry

Industry Level Sentiment

Financial AI impact discourse is **strongly positive** and operationally focused, while AI stock narratives are more volatile, reflecting market-driven uncertainty rather than sustained impact skepticism.

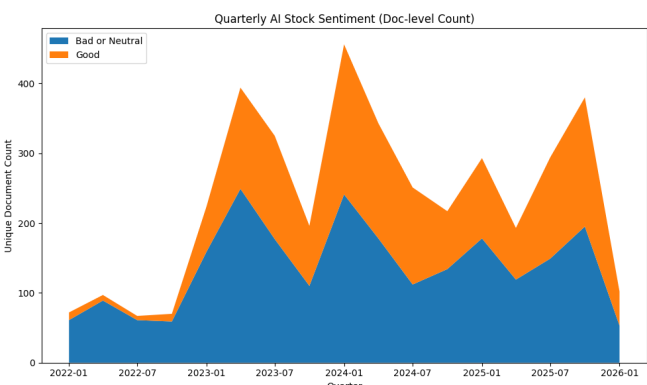
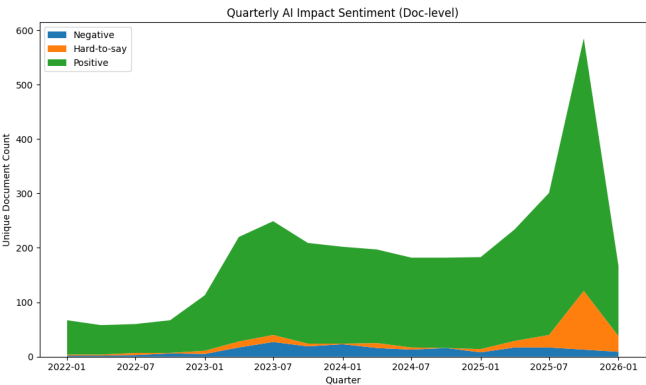


Topic – Level Sentiment

Topic	Sent Score	Ad. Sent Score
AI enhances compliance and lending processes in financial institutions	0.89	0.81
automated trading advisors (EAs) that use AI and logic to trade forex and other pairs	0.72	0.72
how insurers are adopting AI to improve underwriting, claims processing, and operational efficiency	0.95	0.91

Time Level Sentiment

- AI impact discourse in the financial sector is overwhelmingly positive and expands sharply over time, with only limited sustained negative narratives.
- AI stock sentiment is more volatile and cyclical, reflecting market-driven fluctuations rather than a consistently optimistic impact narrative.



NER in Positive content

- Finance companies: JPMorgan Chase, Mastercard, Visa, Bank of America, Goldman Sachs
- AI companies: OpenAI, Google, Microsoft, Amazon, NVIDIA, IBM, Salesforce
- AI Technologies: machine learning, generative AI, large language models, neural networks, natural language processing
- Positive mechanisms: underwriting automation, document processing, fraud detection, chatbots, productivity improvement, cost reduction, risk management, algorithmic trading support

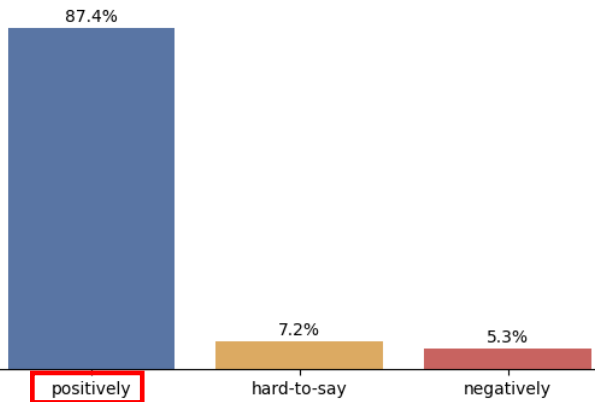
NER in Negative content

- Finance companies: Securities and Exchange Commission, Internal Revenue Service, Federal Reserve, Consumer Financial Protection Bureau, RealPage, Financial Stability Oversight Council
- Negative mechanisms: inaccurate decision-making, job displacement, privacy concerns, loss of trust, bias, risk of using AI-generated content, overreliance on AI, market manipulation risk, litigation risk, system failure

Industry Analysis Result – HealthCare Systems Industry

Industry Level Sentiment

Healthcare AI discourse is **strongly optimistic overall**, driven by **clinical efficiency and innovation narratives**, while **concerns** around safety, bias, and regulation emerge gradually over time.

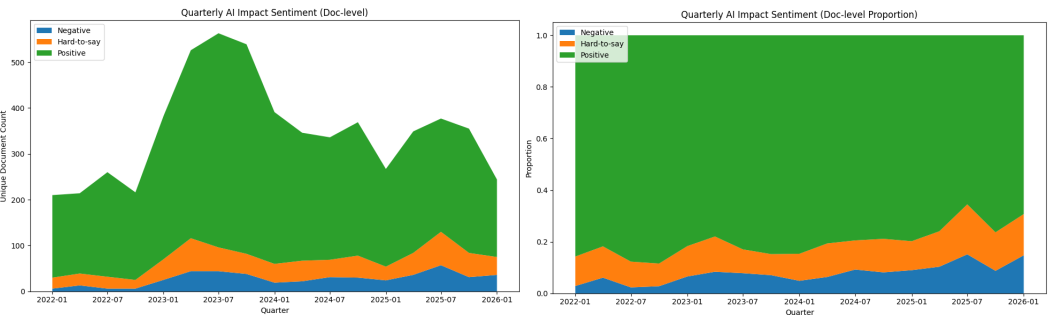


Topic – Level Sentiment

Topic	Sent Score	Ad. Sent Score
AI in Clinical Diagnostics and Imaging	0.91	0.88
AI in Clinical Workflow and Administrative Support	0.96	0.94
AI in Drug Discovery and Biomedical Research	0.89	0.86
AI in Patient Interaction and Education	0.61	0.43
Healthcare AI Market, Regulation, and Ethics	0.70	0.59

Time Level Sentiment

AI impact discourse remains predominantly positive over time, though uncertainty and negative signals gradually increase in later periods.



NER in Positive content

- HealthCare companies: FDA, Mayo Clinic, NHS, GE Healthcare, Philips, Epic, Merck, Siemens Healthineers
- AI companies: DeepMind, Insilico Medicine, Google, Microsoft, OpenAI, Nvidia
- AI Technologies: generative AI, large language models, deep learning, natural language processing, AlphaFold, computer vision, reinforcement learning, federated learning
- Positive mechanisms: drug discovery, workflow automation, operational efficiency, diagnostic support, cost reduction, precision medicine, productivity improvement, clinical decision support, predictive analytics

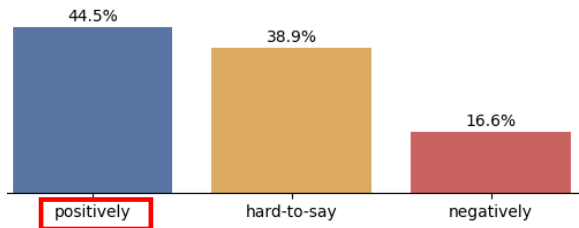
NER in Negative content

- HealthCare companies: Mayo Clinic, Kaiser Permanente, Mass General Brigham, Epic, Pfizer, Humana
- AI Technologies: ambient voice technology, AI scribes, diagnostic AI, health apps, clinical decision-making support
- Negative mechanisms: diagnostic errors, algorithmic bias, patient safety risks, unsafe clinical recommendations, overreliance on AI, loss of trust in healthcare systems, privacy violations, healthcare inequity, regulatory scrutiny, failed AI deployment

Industry Analysis Result – Education and Research Industry

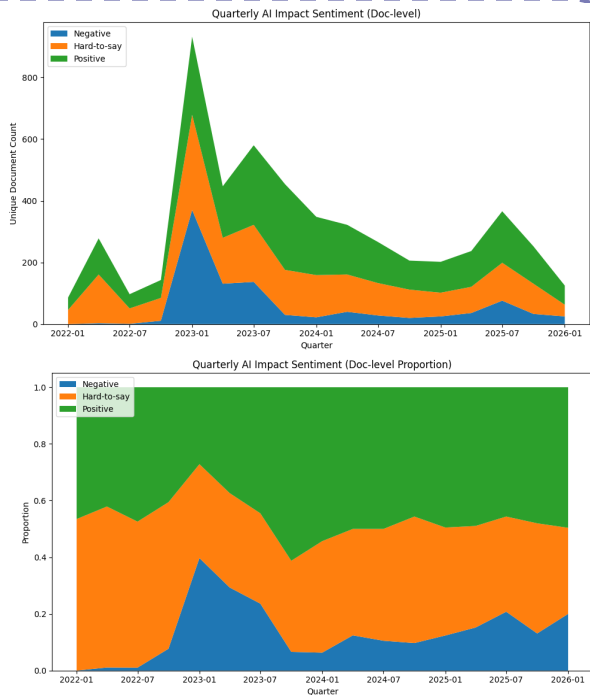
Industry Level Sentiment

AI discourse in education and research is **moderately positive** but highly contested, marked by strong **innovation** narratives alongside persistent concerns over **academic integrity and learning quality**.



Time Level Sentiment

- AI impact discourse shows a sharp surge around early-2023 driven by uncertainty and negative signals,
- followed by stabilization with a persistently dominant but gradually moderating positive sentiment.



In time analysis plot: ● Positive, ● Hard to say, ● Negative

Topic – Level Sentiment

Topic	Sent Score	Ad. Sent Score
Educators are adapting teaching methods to address student cheating with AI tools like ChatGPT	0.41	0.15
Online professional certifications for data science and cloud engineering careers	0.28	0.28
Astronomy and space science news, including discoveries about Mars, stars, and galaxies	0.68	0.68
Using machine learning to predict suicide risk from psychological and neural data.	0.82	0.82

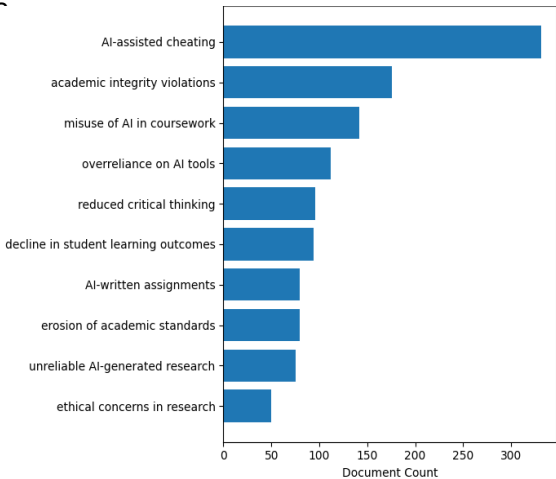
NER in Positive content

- Educational and research institutes: Universities
- AI companies: OpenAI, Google, Microsoft, Amazon
- Positive mechanisms: personalized learning, AI tutoring, automated grading, academic writing assistance, research automation, scientific discovery acceleration, literature review automation, coding assistance for research, knowledge discovery, adaptive learning

NER in Negative content

- Educational institutes: Universities, Public Schools
- AI companies: Turnitin, GPTZero, Hugging Face, Anthropic, Midjourney
- AI Technologies: ChatGPT, AI-assisted cheating, generative AI, AI detection tools, machine learning, AI writing tools

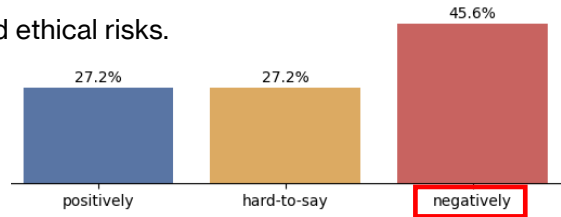
Negative mechanisms:



Industry Analysis Result – Media and Creative Industry

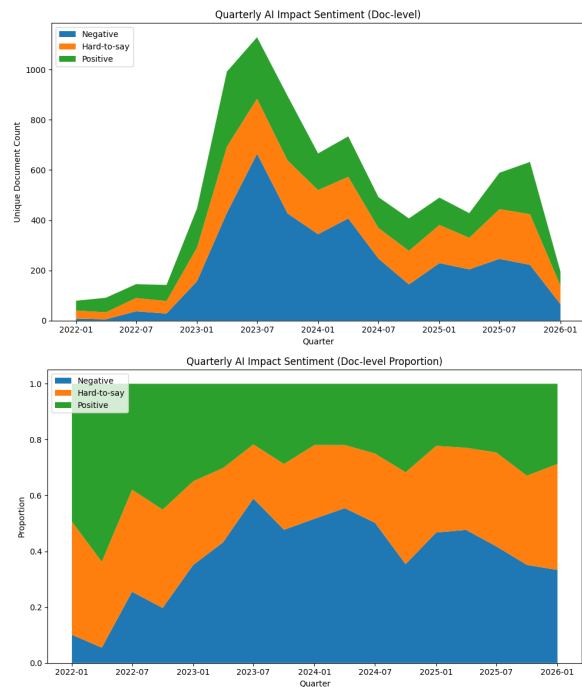
Industry Level Sentiment

AI discourse in the media and creative industry is increasingly **negative-skewed after 2023**, driven by copyright, labor, and authenticity concerns, while positive innovation narratives persist but weaken relative to regulatory and ethical risks.



Time Level Sentiment

- Negative sentiment becomes dominant after 2023,
- while positive narratives weaken and uncertainty remains persistently high



In time analysis plot: ● Positive, ● Hard to say, ● Negative

Topic – Level Sentiment

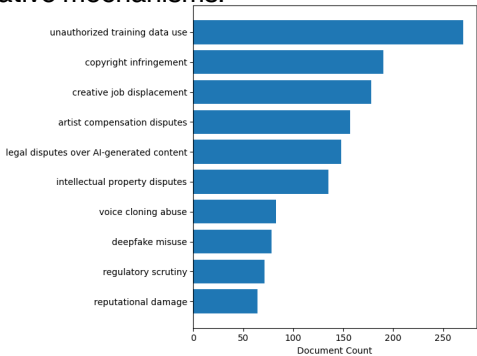
Topic	Sent Score	Ad. Sent Score
Creative Applications and Industry Integration of AI	0.43	0.29
Labor and Ethical Concerns in AI Media Production	0.19	-0.31
Creative Applications and Industry Integration of AI	0.26	-0.27

NER in Positive content

- Media and content creating companies: Spotify, YouTube, Disney, Netflix, TikTok, BBC, Universal Music Group, The New York Times
- AI companies: OpenAI, Google, Microsoft, Suno
- AI Technologies: generative AI, machine learning, large language models, diffusion models, speech synthesis, computer vision, reinforcement learning, evolutionary algorithms,
- Positive mechanisms: music generation, faster content creation, voice cloning, creative workflow automation, cost reduction, personalized media experiences, text-to-image generation, automated content production, recommendation systems

NER in Negative content

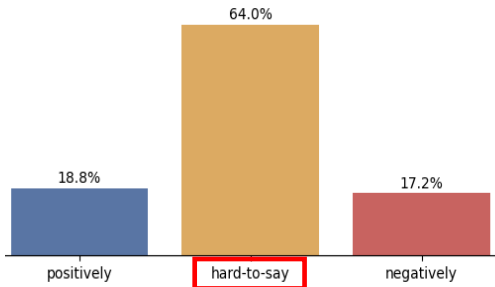
- AI Technologies: voice synthesis, deepfake technology, Stable Diffusion, Midjourney, stream manipulation, regenerative AI, hallucinations
- Negative mechanisms:



Industry Analysis Result – Public Sector Industry

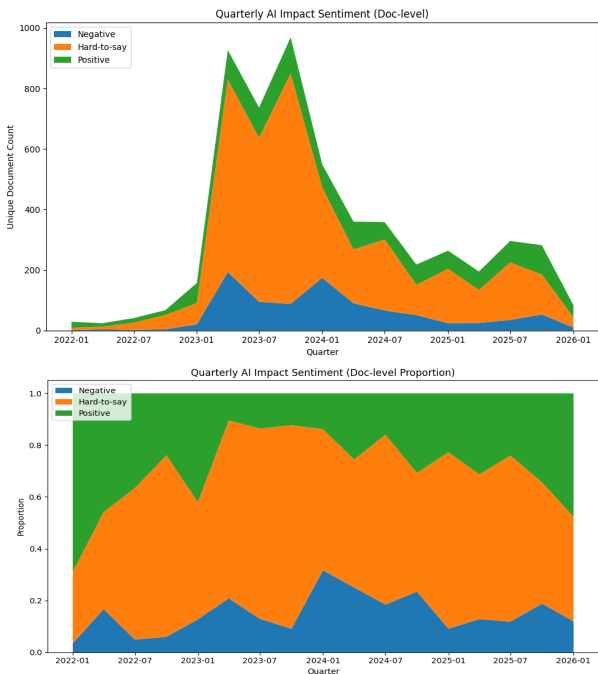
Industry Level Sentiment

Public-sector AI discourse is dominated by **neutral narratives**, with episodic negative spikes around major policy or misinformation events while positive efficiency narratives remain secondary.



Time Level Sentiment

- AI impact discourse is **dominated by uncertainty across most periods**,
- with negative signals episodically rising around major AI waves,
- while positive narratives remain secondary and cyclical.



Topic – Level Sentiment

Topic	Sent Score	Ad. Sent Score
Senate Majority Leader Chuck Schumer is leading bipartisan efforts to craft legislation regulating artificial intelligence.	0.13	0.09
European Union's new AI regulations focus on managing high-risk systems and protecting fundamental rights.	0.07	0.06
Legal technology and AI tools transforming legal research and practice for law firms.	0.58	0.35
Regulate AI-generated deepfakes in political campaign ads to combat election misinformation	0.02	-0.86
The UK's AI Safety Summit at Bletchley Park focused on international cooperation to manage AI risks	0.17	0.16
UN Secretary-General Guterres advise on global AI governance.	0.02	0.01

NER in Positive content

- Public Sectors : national governments, supranational regulators, and public-sector policy institutions
- AI Technologies: generative AI, natural language processing, predictive analytics, synthetic media, foundation models, computer vision, explainable AI
- Positive mechanisms: legal research automation, administrative efficiency, regulatory oversight, compliance monitoring, policy enforcement, public service automation, judicial decision support, workflow automation, misinformation detection, election monitoring

NER in Negative content

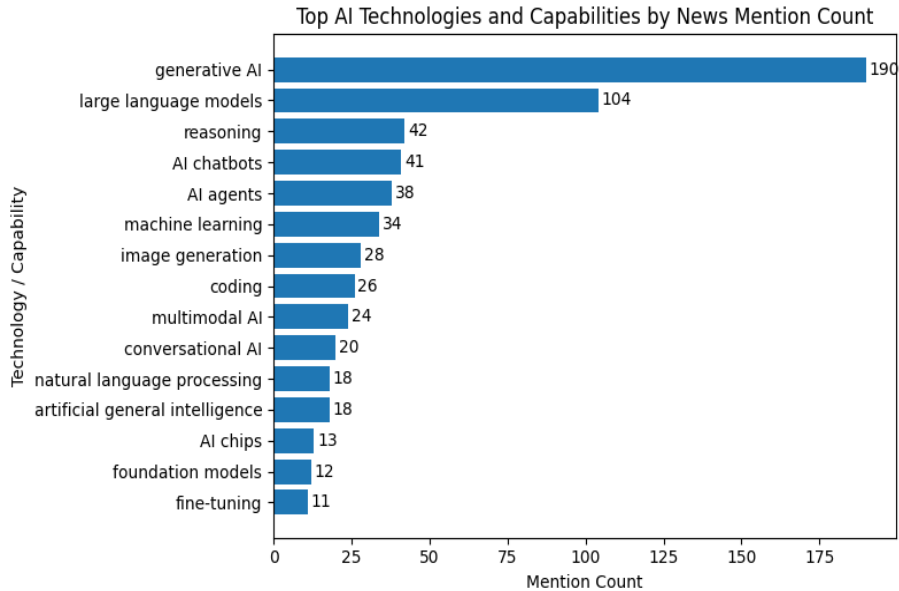
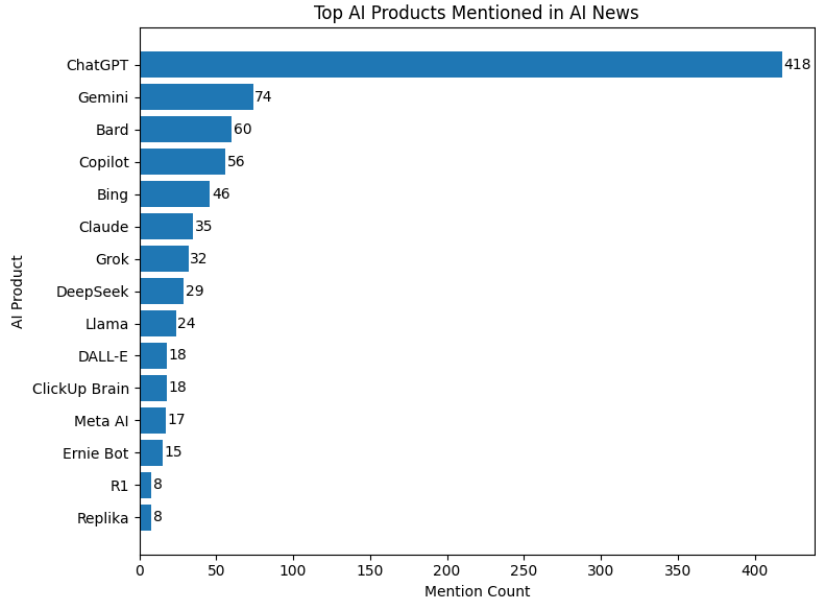
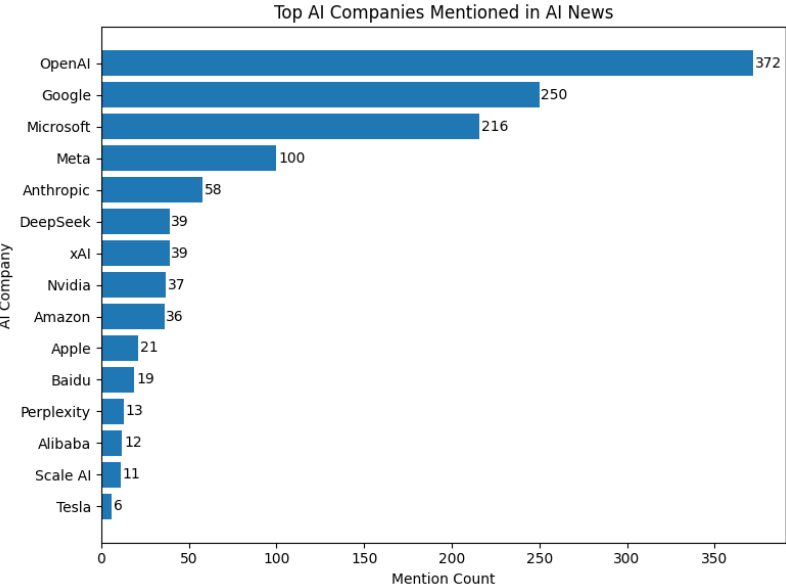
- AI Technologies: deepfakes, AI-generated content, AI tools, hallucinations, synthetic media
- Negative mechanisms: election misinformation/disinformation, legal hallucinations/errors, public trust erosion, compliance failures, regulatory gaps, institutional misuse, bias in public decision-making, administrative disruption, systemic risk, governance failures

Industry Analysis Result – Digital Service Industry

Digital service acts primarily as an **AI innovation driver rather than a passive impact recipient**, so instead of sentiment analysis, the focus is on **what is being discussed within the industry** shaping the sector’s competitive and innovation landscape.

- Dominant topics and strategic themes
- leading AI companies (e.g., OpenAI, Google, Microsoft)
- key AI products (e.g., ChatGPT, Gemini, Copilot)
- The most prominent technologies and capabilities (e.g., generative AI, large language models, reasoning, multimodal AI)

Topic
AI SaaS tools & enterprise productivity platforms
Consumer AI companionship & conversational agents
AI governance & OpenAI corporate dynamics
AI search engine competition
Chinese LLM ecosystem & global AI competition
Microsoft Copilot & AI OS integration
Google Gemini model ecosystem
Google Bard → conversational AI transition
AI safety, misuse & controversial chatbot behavior



Interpretive Analysis – Success & Barriers to AI Adoption

The dissemination of AI impact

It generally follows three structural pathways:

1. Technology / Compute-driven mechanisms
2. Knowledge / Cognitive-driven mechanisms
3. Physical / Operational-driven mechanisms

A Unifying Pattern across industries

Across sectors, AI adoption success does not primarily depend on technological maturity, but on how risk, responsibility, and uncertainty are institutionally structured.

- **AI adoption success is highest where:**
 - task structure is standardized and modularized
 - performance metrics are measurable
 - accountability exposure is limited or diffused
 - trust costs of machine-mediated decisions are low
- **AI adoption resistance is highest where:**
 - decisions carry high social or legal stakes
 - institutional legitimacy and professional authority matter
 - outcomes are embedded in complex human relationships
 - uncertainty cannot be easily externalized to systems

Thus, AI adoption is fundamentally an institutional risk allocation problem, rather than a purely technological challenge.

Strategic Insights for AI Adoption

For firms

- prioritize AI deployment in modular, repeatable workflow segments
- design hybrid human-AI governance rather than full automation
- Invest in measurement infrastructure before AI implementation
- treat AI adoption as organizational change, not IT upgrade

For policymakers

- develop legitimacy frameworks for AI-mediated decisions
- create accountability regimes aligned with AI capability limits
- support institutional experimentation rather than uniform regulation
- distinguish automation risk from augmentation opportunity

For industry evolution

- AI diffusion will follow **institutional readiness gradients**, not technological readiness
- sectors with high legitimacy constraints will experience slower but deeper transformation
- early adopters shape narrative legitimacy that affects cross-industry diffusion

Thanks

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